

IN THIS CHAPTER

- » Looking at the various elements that make up a typical network infrastructure
- » Considering how standards and protocols are used in networking
- » Taking a look at network topology
- » Examining the elements of a network's cable infrastructure
- » Understanding ports, interfaces, and MAC addresses
- » Seeing how network data is transmitted via packets
- » Understanding collisions in wired and wireless networks
- » Introducing broadcast packets
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Chapter 2

Network Infrastructure

In this chapter, I cover the key concepts of local area networks (LANs) — that is, networks that are contained within a single location. Although this chapter may seem a little abstract, you'll be much better prepared to design and implement a solid LAN if you have a good understanding of these concepts from the very beginning.

I go into more depth on many of the concepts presented in this chapter in Book 2, which dives deeper into the various networking standards and protocols.

Introducing Infrastructure

As I mention in the preceding chapter, a *local area network* is a network that connects computers and other devices that are located in relatively close proximity to one another. Most LANs are contained to a single building, although it's possible to create LANs that span several buildings at a single site, provided the buildings are close to one another. For the purposes of this chapter, I stick to LANs that operate within a single building and support anywhere from a few dozen to a few hundred users.

LANs exist to connect computing devices — such as workstation computers, servers, printers, scanners, cameras, and so on — to one another. The essence of a network is the *physical infrastructure* that enables the connections. The infrastructure is similar to the infrastructure of a city. A city's infrastructure has many physical elements, including roads, stop signs and stoplights, water supply lines, stormwater drains, sewage lines and treatment plants, electrical distribution cables, transformers, and much more.

Similarly, the infrastructure of a network consists of physical elements:

- » **Cables:** These run through walls and ceiling spaces, through conduits, between floors, and wherever else they need to go to reach their destinations.
- » **Patch panels:** These allow cables to be organized at a central location.
- » **Network switches:** A *switch* is an intermediate device that sits between the networked devices that allows those devices to communicate with each other. In a real way, switches are the core of the network; without switches, computers wouldn't be able to talk.
- » **Wireless access points (WAPs):** A *wireless access point* lets devices connect wirelessly to the network. Depending on the size of your network and the physical space your users occupy, you may need more than one WAP. Each WAP needs to be connected to the LAN via a cabled switch connection.
- » **At least one router:** A *router* connects the network to the outside world. The most common use of a router is to connect the LAN to the internet. However, routers can also be used to connect one LAN to another. (You can find more about routers in Book 1, Chapter 3.)

Understanding Network Protocols and Standards

To operate efficiently, the infrastructure of a network consists of devices that conform to well-known standards and protocols. A *protocol* provides a precise sequence of steps that each element of a network must follow to enable communications. Protocols also define the precise format of all data that is exchanged in a network. For example, the Internet Protocol (IP) defines the format of IP addresses: four 8-bit numbers called *octets*, whose decimal values range from 0 to 255, as in 10.0.101.155 (see “Perusing Ports, Interfaces, and MAC Addresses,” later in this chapter for more on octets).

A *standard* is a detailed definition of a protocol that has been established by a standards organization and that vendors follow when they create products. Without standards, it would be impossible for one vendor’s products to work with another vendor’s products. Because of standards, you can purchase equipment from different vendors with the assurance that they’ll work together.

Network standards are organized into a framework called the Open Systems Interconnection (OSI) reference model. The OSI model establishes a hierarchy for protocols so that each protocol can deal with just one part of the overall task of data communications. Table 2-1 shows seven distinct layers at which a protocol may operate as described by the OSI reference model.

TABLE 2-1 The Seven Layers of the OSI Model

Layer	Name	Description
1	Physical	Governs the layout of cables and devices, such as repeaters and hubs.
2	Data link	Provides media access control (MAC) addresses to uniquely identify network nodes and a means for data to be sent over the physical layer in the form of packets. Bridges and switches are layer-2 devices.
3	Network	Handles routing of data across network segments.
4	Transport	Provides for reliable delivery of packets.
5	Session	Establishes sessions between network applications.
6	Presentation	Converts data so that systems that use different data formats can exchange information.
7	Application	Allows applications to request network services.

Although the upper layers of the OSI model (layers 4 through 7) are equally important, in this chapter and the next, I focus on the first three layers of the OSI model — physical, data link, and network. These layers are the ones where the most common types of networking hardware (such as cables, interfaces, switches, and routers) operate.

Although many different network protocols and standards can be used in various layers of the OSI model, the most common standard found at layers 1 and 2 is Ethernet. Similarly, the most common standard at layer 3 is IP. You can find more about Ethernet and IP in Book 2, Chapters 2 and 3.

Recognizing Network Topology

The term *network topology* refers to the shape of how the computers and other network components are connected to each other. Several different types of network topologies exist, each with advantages and disadvantages.



TIP

In the following discussion of network topologies, I use two important terms:

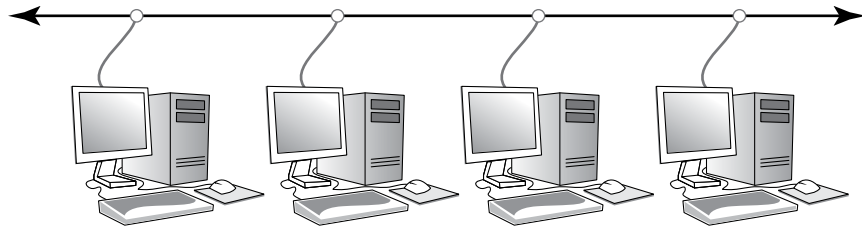
- » **Node:** A *node* is a device that's connected to the network. For your purposes here, a node is the same as a computer. Network topology deals with how the nodes of a network are connected to each other.
- » **Packet:** A *packet* is a message that's sent over the network from one node to another node. The packet includes the address of the node that sent the packet, the address of the node the packet is being sent to, and data.

Bus topology

In a *bus topology*, nodes are strung together in a line, as shown in Figure 2-1. The key to understanding how a bus topology works is to think of the entire network as a single cable, with each node “tapping” into the cable so it can listen in on the packets being sent over that cable.

In a bus topology, every node on the network can see every packet that's sent on the cable. Each node looks at each packet to determine whether the packet is intended for it. If it is, the node claims the packet. If it isn't, the node ignores the packet. This way, each computer can respond to data sent to it and ignore data sent to other computers on the network.

FIGURE 2-1:
Bus topology.



If the cable in a bus network breaks, the entire network is effectively disabled. Obviously, the nodes on opposite sides of the break can't continue to communicate with each other, because data can't span the gap created by the break. But even those nodes that are on the *same* side of the break may not be able to communicate with each other, because the open end of the cable left by the break disrupts the proper transmission of electrical signals.

In the early days of Ethernet networking, bus topology was commonplace. Although, for most networks, bus topology has given way to star topology (see the next section), many networks today still have elements that rely on bus topology.

Star topology

In a *star topology*, each network node is connected to a central device called a *hub* or a *switch*, as shown in Figure 2-2. Star topologies are commonly used with LANs.

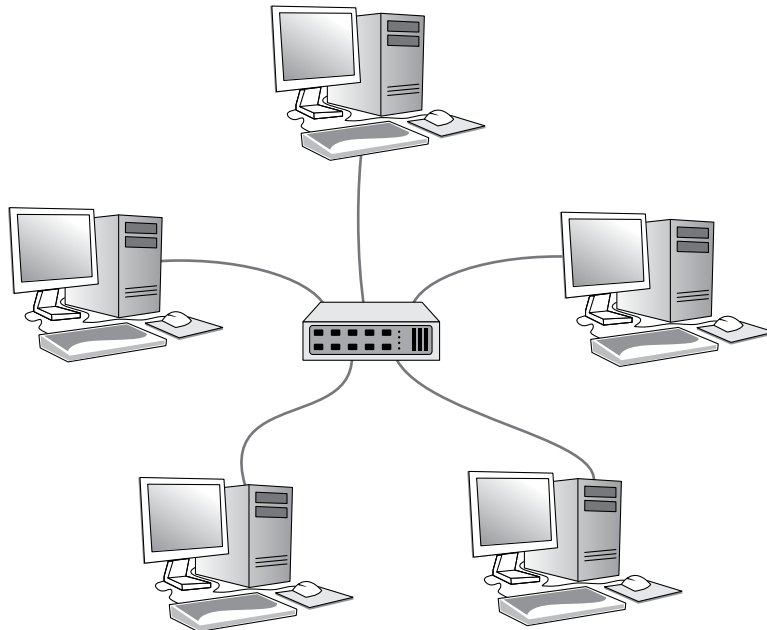


FIGURE 2-2:
Star topology.

If a cable in a star network breaks, only the node connected to that cable is isolated from the network. The other nodes can continue to operate without interruption — unless, of course, the node that's isolated because of the break happens to be the file server.



TECHNICAL
STUFF

You should be aware of the somewhat technical distinction between a hub and a switch. Simply put, a *hub* doesn't know anything about the computers that are connected to each of its ports. So, when a computer connected to the hub sends a packet to a computer that's connected to another port, the hub sends a duplicate copy of the packet to all its ports. In contrast, a switch knows which computer is connected to each of its ports. As a result, when a switch receives a packet intended for a particular computer, it sends the packet only to the port that the recipient is connected to.

Strictly speaking, only networks that use switches have a true star topology. If the network uses a hub, the network topology has the physical appearance of a star, but it's actually a bus. That's because when a hub is used, each computer on the network sees all the packets sent over the network, just as in a bus topology. In a true star topology, as when a switch is used, each computer sees only those packets that were sent specifically to it, as well as packets that were specifically sent to all computers on the network (those types of packets are called *broadcast packets*; see “Dealing with Broadcast Packets,” later in this chapter).

EXPANDING STARS

Physicists say that the universe is expanding, and network administrators know they're right. A simple bus or star topology is suitable only for small networks, with a dozen or so computers. But small networks inevitably become large networks as more computers are added. For larger networks, it's common to create more complicated topologies that combine stars and buses.

For example, a bus can be used to connect several stars. In this case, two or more hubs or switches are connected to each other using a bus. Each of these hubs or switches is then the center of a star that connects two or more computers to the network. This type of arrangement is commonly used in buildings that have two or more distinct workgroups. The bus that connects the switches is sometimes called a *backbone*.

Another way to expand a star topology is to use a technique called *daisy-chaining*. When you use daisy-chaining, a switch is connected to another switch as if it were one of the nodes on the star. Then this second switch serves as the center of a second star.

Ring topology

In a ring topology, packets are sent around the circle from computer to computer, as shown in Figure 2-3. Each computer looks at each packet to decide whether the packet was intended for it. If it isn't, the packet is passed on to the next computer in the ring.

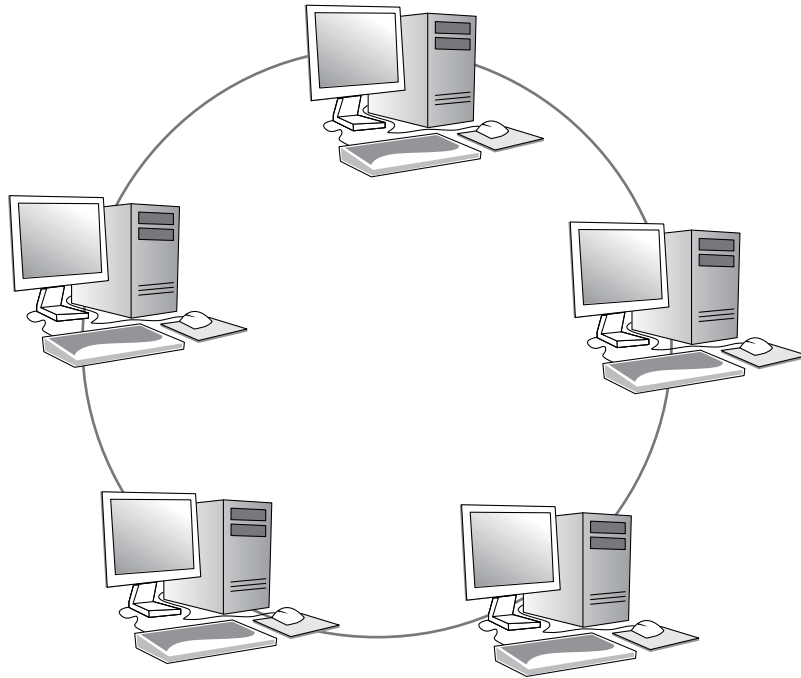


FIGURE 2-3:
Ring topology.

Years ago, ring topologies were common in LANs, because two popular networking technologies used rings: Attached Resource Computer Network (ARCNET) and Token Ring. ARCNET is still used for certain applications, such as factory automation, but it's rarely used in business networks. Token Ring is still a popular network technology for IBM midrange computers. Although plenty of Token Ring networks are still in existence, not many new networks use Token Ring any more.

Ring topology was also used by Fiber Distributed Data Interface (FDDI), one of the first types of fiber-optic network connections. FDDI has given way to more efficient fiber-optic techniques, however. So, ring networks have all but vanished from business networks.

Mesh topology

In a *mesh topology*, multiple connections exist between each of the nodes on the network, as shown in Figure 2-4. The advantage of a mesh topology is that if one cable breaks, the network can use an alternative route to deliver its packets.

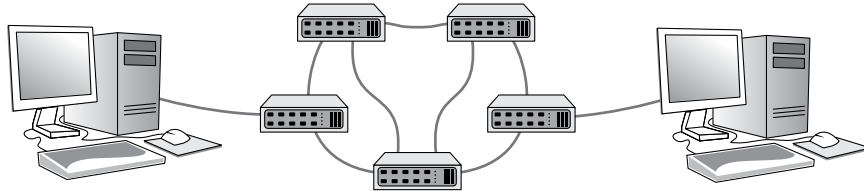


FIGURE 2-4:
Mesh topology.

Mesh networks are often used to link switches in a LAN. In Figure 2-4, the mesh has a total of seven connections. If any of these seven connections goes bad, any switch can still reach any other switch while traveling through, at most, one intermediate switch.

Mesh networks are also very common for metropolitan area networks (MANs) or wide area networks (WANs). These networks use routers to route packets from network to network. For reliability and performance reasons, routers are usually arranged in a way that provides multiple paths between any two nodes on the network in a mesh-like arrangement.

Considering Cable

You can find much more about the details of working with network cable in Book 3, Chapter 1, as well as Book 4, Chapter 1. But before we get too far, I want to give you an overview of what's involved with cabling together a network.

For starters, network cable and all the bits and pieces that go along with it are the most important components of layer 1 of the OSI reference model. The following sections describe the most important layer 1 and cabling details you need to know.

Twisted-pair cable

You can choose from several varieties of cable, but the most common is called *twisted-pair*. It's called that because inside the outer sheath of the cable are four pairs of small insulated wire. The wires are 24 gauge, which means they're about

half a millimeter in diameter. These pairs are color coded: blue, green, orange, and brown. For each pair, there is one solid-colored wire and one striped wire — so, the blue pair consists of a solid blue wire and a blue-and-white striped wire.

The two wires that make up each pair are twisted together in a way that prevents the electrical signals within each pair from interfering with the other pairs. To accomplish this, each pair is twisted at a different rate.

The maximum length of a single run of Cat5e cable is 100 meters.

Cat5e cable is able to carry network data at speeds of up to 1 gigabit per second (Gbps). The newer and somewhat more expensive Cat6 cable can carry data at up to 10 Gbps but can sustain that speed for only 55 meters.

RJ45 connectors

Twisted-pair cable is attached to network devices using a special type of connector called an RJ45, which is a small block of plastic with eight metal contacts. RJ45 connectors resemble telephone connectors, but they're larger (telephone connectors have just four electrical contacts). For the cable to meet Cat5e standards, the twists of the individual pairs must be maintained all the way up to the RJ45 connector.

RJ45 connectors come in both male (plug) and female (receptacle) varieties. Typically, the male connector is installed on the cables and the female connectors are installed in equipment. So, in order to connect a cable to a computer, you plug the male RJ45 plug on the cable into the female RJ45 receptacle on the computer.

Patch panels and patch cables

A *patch panel* is a group of RJ45 receptacles on a single metal plate, usually attached to a 19-inch equipment rack. Patch panels are used to bring cables that are run from individual computer locations to a single location where they can then be patched to other equipment using patch cables. A *patch cable* is simply a short length of twisted-pair cable with an RJ45 plug on both ends. Patch cables are usually 3 to 10 feet in length, but longer lengths are occasionally used.

Patch panels typically have either 24 or 48 ports. Depending on the size of your network, you may have more than one patch panel at a single location. For example, a large network may have four 48-port patch panels to support a total of 192 computers.



A patch panel by itself doesn't actually *do* anything. Its job is simply to provide a central collecting point for all your network cables so that you can easily use patch cables to connect the cables to other devices, such as switches or servers.

Repeaters and hubs

A *repeater* is a layer-1 device that is designed to circumvent the maximum length limitation of twisted-pair network cables. A repeater contains two RJ45 ports, which are connected internally by an amplifier. Electrical signals received on either of the two ports are boosted by the amplifier and sent through the other port. Thus, the cables on both ends of the repeater can be up to 100 meters. The repeater effectively doubles the reach of the cable.

A *hub* is a repeater with more than two ports. For example, a hub may have four or eight ports. Each of these ports can connect to another device on the network, such as a client computer, a server, or a printer. A port on a hub can also connect to another hub, so that (for example) an eight-port hub can connect to seven computers and another eight-port hub, which can connect to seven more computers. In this way, two eight-port hubs can connect 14 computers to each other.

There are two very important things to know about hubs:

- » **An electrical signal received on any of the hub's ports is amplified and repeated on all the other ports in the hub.** So, in an eight-port hub, any electrical signals received on port 1 are amplified and then sent out on ports 2 through 8. Any devices that are connected to ports 2 through 8 see the signals that were received on port 1. The same is true for signals received on any of the other ports; for example, any signals received on port 4 will be amplified and repeated on ports 1 through 3, as well as ports 5 through 8.
- » **Hubs are almost never used anymore.** That's because simply repeating all incoming signals on all ports is an incredibly bad idea, for reasons that will become apparent later in this chapter and in Book 1, Chapter 3. If your network still has hubs, you should seriously consider replacing them with switches, which I describe in the next section and explain further in the next chapter.

Switches

A *switch* is a layer-2 device that is similar to a hub in that it allows you to connect more than one device, and packets received on one port are relayed to other ports. The difference, however, is that a switch is able to examine the actual contents of the data that it receives. As I explain in the "Pondering Packets" section, later in this chapter, data is sent in units called *packets* that contain a destination address.

A switch looks at this destination address and repeats the incoming packet only on the port that can deliver the packet to the intended destination.

For example, suppose computer A is connected to switch port 1, and computer D is connected to switch port 4. If computer A sends a packet to computer D, that packet is received on switch port 1. The switch knows that computer D is connected to switch port 4, so the switch sends the packet out on switch port 4. In this way, computer D receives the packet. The computers or devices that are connected to the other ports on the switch are not bothered with the packet intended for computer D.

If that doesn't make a lot of sense, don't worry — it will. The next two sections in this chapter explain the concept of MAC addresses, which are how networks identify the intended recipients of data packets, as well as how data packets work. Then, in Book 1, Chapter 3, I dive deeper into how switches do their magic.

Perusing Ports, Interfaces, and MAC Addresses

A *network interface* is the electronic circuitry that allows a device to connect to a network. Each network interface provides a *port*, which is the plug-in point for the interface. Generally speaking, the terms *port* and *interface* are synonymous.

A network interface may be a separate add-on card for a computer, in which case the interface is called a *network interface card* (NIC). On some devices, such as printers, separate NICs are still common. But nearly all desktop and laptop computers have a network interface built into the computer's motherboard, so separate NICs are rarely used on desktop computers or laptops. NICs are still widely used on servers, however, because servers are often configured with two or more interfaces; using a separate card for the interface allows for more flexibility.



TIP

The term *adapter* is often used as a synonym for *interface*. *Port*, *interface*, *adapter* — three words that mean the same thing.

Every network interface must have a unique identifier called a *MAC address*. (MAC stands for *media access control*, but that won't be on the test.) Each MAC address is unique throughout the entire world. I have no idea whether MAC addresses are unique throughout the galaxy; it's entirely possible that the computer system on some invading alien spacecraft would have a MAC address that is the same as your laptop, but if that were to happen, I doubt you'd be too concerned about fixing your network.

MAC addresses are important because they provide the means for a network to keep track of the devices that make up the network. Without MAC addresses, it would be impossible to know what devices are on the network. And it would be impossible to send information to a particular device or to know which particular device sent information.



TIP

The term *physical address* is sometimes used as a synonym for *MAC address*. The two terms are interchangeable.



TECHNICAL
STUFF

MAC addresses are a part of layer 2 of the OSI reference model, called the link layer. This layer is responsible for the exchange of basic information on a network. The ability to uniquely identify every device on a network is a key component of enabling that to happen.

MAC addresses are 48 binary bits in length, which means that more than 280 trillion devices can be assigned unique MAC addresses before we run out. When written, MAC addresses are written in a peculiar notation that uses six two-digit hexadecimal numbers (each called an *octet*), separated by hyphens. Hexadecimal notation uses a combination of the ten digits 0 through 9, plus the letters A through F, to represent decimal values from 0 through 15 with a single symbol. For example, the decimal value 10 is represented by the letter A, 11 by the letter B, and so on. A typical MAC address looks like this:

```
48-2C-6A-1E-59-3D
```

The details of binary, hexadecimal, and octets are not important for the purposes of this chapter. I take a deep dive into these subjects in Book 2, Chapter 3.

If you want to see the MAC address of your computer's network adapter, open a command prompt and type **ipconfig /all**. Scroll through the output from this command to see the MAC address (ipconfig calls a physical address) for each interface on your computer. For example, here's the ipconfig output for the built-in adapter on my Microsoft Surface Book:

```
Ethernet adapter Ethernet 2:
    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix . . . : lowewriter.com
    Description . . . . . : Surface Ethernet Adapter
    Physical Address . . . . . : 58-82-A8-9C-A7-28
    DHCP Enabled . . . . . : Yes
    Autoconfiguration Enabled . . . . . : Yes
```

Here, you can see the MAC address is 58-82-A8-9C-A7-28.



TIP

A MAC address is technically associated with a network interface, not with the device that uses that interface. For example, if your computer's motherboard has a network interface built in, the MAC address of the network interface is pretty much married to the motherboard. However, if your computer has a separate NIC, the MAC address is a part of the card, not the computer that the card is plugged into. If you remove the interface card from one computer and install it in another, the MAC address travels with the card.



REMEMBER

The key point to remember here is that in order for a computer, printer, or any other device to connect to a network, that device must contain a network interface. That interface has a unique MAC address, which is the primary way that the network can distinguish one device from another.

Pondering Packets

When two or more devices are connected to a network via cables plugged into their network interfaces, those devices can exchange information with one another. This bit of magic is accomplished through the use of *packets*, which are relatively small units of data that are sent and received through the network interface and cables. A network packet always originates at a single network interface, called the *sender*, and it's usually (but not always) sent to a single network interface, called the *destination*.

A packet is very similar to an envelope that you would send through standard mail delivery. It includes the MAC address of both the sender and the destination, as well as some other interesting header information, along with a *payload* that contains the actual data being sent by the packet. You can think of the payload as what you would put in an envelope you want to send through the mail. You wouldn't dream of dropping an envelope in the mail without writing the recipient's address, as well as your own address, on the envelope. So it is with packets.

The payload of an Ethernet packet may be a packet created by some higher-level protocol, such as IP. This is analogous to putting a letter in an envelope, putting that envelope in a larger envelope, and sending it through the mail. When the recipient receives your mail, they open the envelope only to find another envelope that must be opened. That envelope may itself contain another envelope and so on, like Russian nesting dolls.



The term *frame* is often used instead of *packet*, but technically they're not quite the same. Every packet begins with a *preamble*, which consists of 56 bits of alternating zeros and ones. This preamble is used by the electronic circuitry of the interfaces to get their clocks synchronized properly so they can accurately read the rest of the packet. It's the rest of the packet that is technically called the *frame*. In other words, a *packet* consists of a *preamble* followed by a *frame*. Because the preamble is of concern only to the electronic engineers who design network interfaces, most non-engineers use the terms *packet* and *frame* interchangeably.

Ethernet has a standard packet format that all packets sent on an Ethernet network must follow. An Ethernet packet contains the following information:

- » **Preamble (56 bits):** The preamble consists of alternating ones and zeros and is used to synchronize the precise timing required to read packet data.
- » **Start-of-frame marker (1 byte):** A start-of-frame marker is a single byte that indicates that the frame is about to begin.
- » **Destination MAC address (6 bytes).**
- » **Sender MAC address (6 bytes).**
- » **Tag (4 bytes):** The tag, which is used to support virtual local area networks (VLANs), is optional. A VLAN lets you divide two or more distinct LANs on a shared physical infrastructure (for example, cables and switches). (For more information about VLANs, see Book 1, Chapter 3, as well as Book 3, Chapter 1.)
- » **Ethertype (2 bytes):** This field indicates the specific protocol that is contained in the payload.
- » **Payload (46 to 1,500 bytes):** The payload contains the actual data being sent by the packet. If the information that needs to be sent is longer than 1,500 bytes, the information must be broken into two or more packets, sent separately, and then reassembled when the packets reach their destination. (The tasks of breaking up and reassembling the data are handled by protocols at higher layers in the OSI reference model; Ethernet itself has no understanding of what is in the packets it sends.)
- » **Frame check sequence (4 bytes):** The frame check sequence (FCS) is used to ensure that the frame data was sent correctly. Basically, the interface that sends the packet uses an algorithm to calculate a 4-byte number based on the contents of the frame and saves this number in the FCS field. When the packet is received, the receiving interface repeats the calculation and then makes sure that the number recorded in the FCS portion of the packet matches the number it calculated. If the numbers disagree, the packet got garbled in transmission and is discarded.

Note that the details of an Ethernet packet are not really of much concern when you design and implement a network. Here are the main points to remember:

- » Ethernet packets contain the MAC addresses of the sender and the receiver.
- » The payload of an Ethernet packet is almost always a packet created by another higher-level protocol such as IP.
- » Ethernet packets can contain a tag field used to implement VLANs, which provide an important means of organizing a large network into smaller parts that can be more easily managed.

Contemplating Collisions

One of the basic principles of Ethernet is that multiple devices can be connected to shared media (that is, cables), and that all devices connected to this media can and should examine every packet that is sent on the media. In other words, Ethernet uses shared media. (You'll find more information about this in Book 1, Chapter 3.)

Every packet contains the MAC address of the intended recipient. So, when an interface detects an incoming packet, it inspects the recipient MAC address and compares it with its own MAC address. If the addresses match, the interface passes the packet up to the next higher protocol on the protocol stack (typically, IP). If the addresses don't match, the interface assumes that the packet doesn't belong to the interface, so the interface simply ignores the packet.

Using hubs on an Ethernet propagates the shared cable through the network. That's because a hub simply amplifies any packet that arrives on any of its ports and then forwards the amplified packet to all the other ports in the hub. So, if you use a 12-port hub to connect 12 computers together, all 12 of the computers will see all the packets generated by any of the other computers. And if two or more of the computers try to transmit a packet at the same time, the packets will collide.

Ethernet has been very successful — in fact, it has become one of the most widely used networking protocols of all time. However, Ethernet's shared media approach has a basic problem: It doesn't scale well. When two or more interfaces are shared on a single cable, there is always the possibility that two or more interfaces will try to send information at the same time. This is called a *collision*. The result of a collision between two packets is that both packets will be destroyed in the process and will need to be sent again.

In a small network with just a few computers, collisions happen now and again but aren't a big deal. However, in a large network with dozens or hundreds of devices, collisions can become a constant annoyance. In fact, collisions can become such a problem that the network slows to a halt and no one is able to get anything done.



TIP

As a result, it's important to design a network in a way that reduces the possibility of collisions becoming a problem. Fortunately, that's easy to do with modern network equipment: All you have to do is use switches instead of hubs. Switches all but eliminate the problem of collisions by forwarding network packets only to the cable segments that the destination devices are connected to rather than forwarding them throughout the entire network.

You can find out more about how switches work and why it's so important in Book 1, Chapter 3.

Dealing with Broadcast Packets

Not all packets on an Ethernet network are intended for a single destination. Instead, some packets, called *broadcast packets*, are intended to be received by every device on the network. To send a broadcast packet, the sending interface sets the destination MAC address to FF-FF-FF-FF-FF-FF, which is the largest possible MAC address. Then all interfaces that receive the packet inspect the destination, see that the packet is a broadcast packet, and pass the packet up to the next higher protocol.

One of the most common users of broadcast packets is Dynamic Host Configuration Protocol (DHCP), which allows computers that join a network to be assigned an IP address. When a network interface is first connected to a network, it sends out a broadcast packet requesting the address of the network's DHCP server. Every device on the network sees this packet. But only the DHCP service will respond.



WARNING

As you see in the next chapter, broadcast packets can sometimes cause serious problems on your network. All networks should be planned in a way that minimizes problems caused by broadcast packets.

Examining Wireless Networks

As I mention in Book 1, Chapter 1, a *wireless network* is a network in which radio signals are used to connect devices to the network instead of using physical cables. You learn much more about wireless networks in Book 2, Chapter 1, as well as Book 4, Chapter 2. But for now, I want you to keep the following points in mind:

- » **Just like with a wired network, a device connecting to a wireless network does so via a network interface.** A wireless interface, also known as a wireless adapter, includes a radio transmitter and receiver rather than a physical cable connection.
- » **Every wireless network adapter has a MAC address.**
- » **Instead of connecting to a switch or a hub, wireless devices connect to a WAP.**
- » **Collisions are likely on a WAP, just as they are on a hub.** Unfortunately, there is no equivalent to a wireless switch that reduces the collision problem. WAPs are essentially hubs in that every device that connects to the WAP is competing for the same bandwidth. Whenever the WAP sends a packet, all devices connected to the WAP must inspect the packet to determine the MAC address destination. And if two devices try to send packets at the same time, a collision will occur. This is one of the inherent reasons that wireless networking is slower than wired networking.

